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Crosscutting Areas

Robust Contract Designs: Linear Contracts and Moral Hazard

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Abstract. We consider incentive compensation where the firm has ambiguity on the effort-contingent output distribution: The parameters of the output probability distribution are in an ellipsoidal uncertainty set. The firm evaluates any contract by its worst-case performance over all possible parameters in the uncertainty set. Similarly, the incentive compatible condition for the agent must hold for all possible parameters in the uncertainty set. The firm is financially risk neutral and the agent has limited liability. We find that when the agent is financially risk neutral, the optimal robust contract is a linear contract—paying the agent a base payment and a fixed share of the output. Moreover, the linear contract is the only type of contract that is robust to the parameter uncertainty. When there is model uncertainty over a general effort-contingent output distribution, we show that a generalized linear contract is uniquely optimal. When the agent is risk averse and has a piecewise linear utility, the only optimal contract is a piecewise linear contract that consists of progressive fixed payments and linear rewards with progressive commission rates. We also provide the analysis for the trade-off between robustness and worst-case performance and show that our results are robust to a variety of settings, including cases with general ℓ^p -norm uncertainty sets, multiple effort levels, and so on. Our paper provides a new explanation for the popularity of linear contracts and piecewise linear contracts in practice and introduces a flexible modeling approach for robust contract designs with model uncertainty.

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1. Introduction

Incentive contracts are pervasive in modern business, particularly in agency relationships. Firms design a variety of incentive contracts to resolve conflicts of interest between the agent and the principal (Prendergast 1999). Among various types of contracts, simple and linear ones are popular in practice (Holmström 2017). A typical example is salesforce compensation. It is reported that 24% of surveyed companies use commissions only (i.e., rewards proportional to sales), and 35% use both commissions and bonuses to pay their salespeople (Joseph and Kalwani 1998). Other evidence includes chief executive officer (CEO) compensation, where a percentage of targeted bonuses is paid to the CEO by public companies (Murphy 2000); franchise contracts, where a franchisee is required to share a percentage of revenues (i.e., royalties) with a franchisor (Bhattacharyya and Lafontaine 1995); and so on. The intriguing question is how to explain the popularity of linear contracts in practice.

Agency theory provides foundations to design effective incentive contracts (Laffont and Martimort 2009). However, the existing literature typically shows that optimal incentive contracts are complex and non-linear (Oyer 2000). This raises a fundamental question in agency theory, particularly in salesforce compensation: why linear- or commission-based contracts are so common in practice, whereas classical agency theory often prescribes more complicated contract forms (Conlon 2009)? This is an outstanding question in salesforce incentives and agency theory. As Holmström and Milgrom (1987, p. 326) state in their classic paper on linear contracts in dynamic environments:

It is probably the great robustness of linear rules based on aggregates that accounts for their popularity. That point is not made as effectively as we would like by our model; we suspect that it cannot be made effectively in any traditional Bayesian model. But issues of robustness lie at the heart of explaining any incentive scheme which is expected to work well in practical environments.

This paper aims to offer an alternative explanation to this problem by incorporating the classical agency theory/moral hazard with the consideration of parameter/model uncertainty. Model uncertainty refers to decision makers' ambiguity or uncertainty over the distributional models (Hansen and Sargent 2008). It is ubiquitous as in practice only limited historical data are available, instead of distributional information. With limited data, the statistical inference is subject to model/parameter ambiguity. For instance, when firms forecast future demand distributions using historical sales data, they usually obtain confidence regions for parameter estimates of the distributions. This issue can be exacerbated when the historical data are not available, for example, new products or new agents, or incomplete. As another example, consider a manufacturer enters a new market by employing a local retailer, where the latter possesses more information or data about consumers than the former.¹ For example, Walmart collects abundant sales data about consumers, and the data Walmart collected are more than twice of those in the internet (Hays 2004). Ideally, the manufacturer would like to design a supply contract that is robust to model uncertainty so that the local retailer can always behave. With model uncertainty, the traditional incentive contracts based on the classical agency theory may fail to incentivize the agent as it requires complete model information. It is unclear how to design incentive contracts to incentivize agents in the presence of model uncertainty. What should be the form of such a contract? Would a simple linear contract be optimal?

In our model, as in standard moral hazard, the agent exerts an unobservable costly effort, which affects the random output. The firm provides incentives by paying the agent based on the observed output. The objective of the risk-neutral firm is to maximize the expected value of output minus the wage paid out. The agent has limited liability. In contrast to most of the existing literature, we consider that the firm does not exactly know the parameters of the effort-contingent output distribution, but knows that the parameters can be in some range, for example, an ellipsoidal uncertainty set. In the presence of the parameter uncertainty, we propose the notion of *robust contract design*: (1) any contract is judged by its worst-case performance, given the firm's knowledge; (2) the incentive compatibility condition holds for any possible parameters in the uncertainty set, for example, the firm would like the agent to exert high effort for any possible parameters in the uncertainty set; and (3) the firm would like to identify the contract that maximizes its own worst-case performance.

We find that when the agent is financially risk neutral, the optimal robust contract is a linear contract—paying the agent a base payment and a fixed share or

commission of the realized output. In fact, it is the only type of contract that is robust to the parameter uncertainty. In other words, nonlinear contracts may fail to incentivize the agent to exert high effort if there is ambiguity over the output distribution. Therefore, our result partially answers the call from Holmström and Milgrom (1987) as our contract is non-Bayesian and the parameter uncertainty can be the driving factor. Furthermore, when the agent is financially risk averse and has a piecewise linear concave utility, we find that the only optimal robust contract is a piecewise linear contract that consists of progressive fixed payments and linear rewards with progressive commission rates. This type of contract is not uncommon in practice. For example, Murphy (2000) reports that among 177 public companies in the United States, 80% use capped bonuses and multiple performance thresholds in their CEO compensation. Thus, our findings also provide an alternative explanation for commonly used piecewise linear contracts: the robustness of the piecewise linear contracts accounts for their prevalence in practice.

There is also an inherent trade-off between the robustness and the worst-case performance: with a larger uncertainty set, the firm is more confident while it has to pay a higher commission rate to the agent. This implies that the parameter uncertainty benefits the agent. As a result, the agent has incentive to inflate the parameter uncertainty if the firm relies on the agent's information. If the firm can form a confidence level for the uncertainty set, we find that the optimal choice of the confidence level is decreasing in the possible outputs. Hence, when the possible outputs are higher, the firm should obtain more accurate market information on the parameters to reduce model uncertainty.

What is the intuition that a simple linear contract is robust to the parameter uncertainty? Note that with the parameter uncertainty, the firm's original risk-neutral utility becomes a risk-averse surrogate utility under the worst-case criterion, which consists of a risk-neutral utility minus a uncertainty-adjusted (or robustness loss) term. The uncertainty-adjusted term is in the form of a norm of the possible outputs. Similarly, under the robust incentive compatibility, the agent's original risk-neutral utility becomes a risk-averse surrogate utility with a homogenous uncertainty-adjusted term. To share the output between these two parties with norm-form risk-averse surrogate utilities, a linear- or commission-based contract is required. Essentially, with the parameter uncertainty, to incentivize the agent, a quota-based bonus payment is not sufficient as both the firm and the agent exhibit a risk-aversion property with uncertainty-adjusted terms.

Our results are robust to a variety of settings. We investigate the case where the firm has more general

model uncertainty instead of the parameter uncertainty, that is, the firm has ambiguity over a general effort-contingent output distribution with the monotone likelihood ratio property (MLRP). We show that a generalized linear contract is uniquely optimal in the presence of model uncertainty, in which the firm does not reward the agent when the agent's performance is below some quota (threshold), but otherwise rewards the agent proportionally in addition to a base payment. The linear part is again due to the robustness, whereas the advent of the threshold is owing to the MLRP. We also show that our results are robust to cases with ℓ^p -norm uncertainty sets, multiple effort levels, and the mean-variance agent utility. The linear contract remains optimal under the robust participation constraint, and inconsistent information sets (see Online Appendix C). In particular, the robust participation constraint does not affect the optimal contract structurally, but changes the base payment. In summary, the norm-formed uncertainty set and the worst-case criterion are the driving factors for the robustness of (piecewise) linear contracts.

We contribute to the extant literature on incentive contracts and agency theory in the following three aspects:

- We provide a new explanation and micro-foundation on why linear-based contracts and quota-based piecewise linear contracts are so common in practice, particularly in salesforce and CEO compensations, that is, parameter uncertainty or model uncertainty and the worst-case criterion can be the driving factors of the popularity of linear rules due to their robustness.
- As the optimal robust contract is simple and intuitive, we provide guidelines and insights on how to design incentive contracts in practice. For new products or agents, firms usually have significant model uncertainty and a commission-based or quota- (progressive) commission-based compensation is strongly recommended.
- We provide a modeling framework and new tools for analyzing agency problems and contract designs with analytically trackable methods to deal with model uncertainty. In particular, we illustrate how to show the optimality of (piecewise) linear contracts through a novel way by exploiting the norm-form adjusted terms to yield simple proofs and solutions. In addition, we provide the trade-off between robustness and worst-case performance.

The rest of the paper is organized as follows. We provide the related literature in Section 2. The model is presented in Section 3. In Sections 4 and 5, we provide our main results and the robustness checks, respectively. Finally, we provide concluding remarks in Section 6. The main proofs can be found in Online

Appendix A. Other additional analyses and proofs are presented in Online Appendices B and C.

2. Related Literature

Our work directly contributes to the classical agency theory built on the principal-agent moral hazard model. This stream of literature is vast. We only review the related literature from the solution perspective (see Laffont and Martimort 2009 for a comprehensive treatment). Grossman and Hart (1983) validate the first-order approach (FOA) for general discrete principal-agent problems. They transform the contract design problem into a convex optimization problem by treating the agent's utility as decision variables. The validation of the FOA for continuous principal-agent problems can be found in Conlon (2009), Kirkegaard (2017), and so on. However, in this literature, to validate the FOA, we require restrictive conditions, such as the MLRP, and the concavity of the distribution function (CDFC) (Rogerson 1985). We adopt the discrete model similar to Grossman and Hart (1983) for tractability. In contrast to the literature cited previously, which assumes complete model information, we assume the principal is uncertain about the output distribution. In this sense, our model enriches the model in Grossman and Hart (1983) with model uncertainty.

Our work also contributes to the salesforce compensation literature pioneered by Basu et al. (1985), who apply the agency theory to investigate the salesforce incentives. This stream of literature mainly focuses on two types of contracts: quota-bonus contracts (see, e.g., Park 1995, Kim 1997) and commission-based or linear contracts (see, e.g., Rao 1990, Lal and Srinivasan 1993), both of which are quite popular in practice. Oyer (2000), Chu and Lai (2013), and Dai and Jerath (2013, 2016) show that the quota-bonus contract, which is nonlinear, is (weakly) optimal in a variety of settings such as limited liability, rent sharing, and limited inventory. Holmström and Milgrom (1987) show the optimality of linear contracts to the moral hazard problem and Laffont and Tirole (1986), Rogerson (2003), and Chu and Sappington (2007) suggest a menu of linear contracts considering private cost information. Inspired by Holmström and Milgrom (1987), we show that the (piecewise) linear contract is the only optimal contract for salesforce compensation, or a general agency problem subject to model uncertainty.

Our work is also related to the literature on the optimality of linear contracts with incomplete knowledge of uncertain environments. Diamond (1998) establishes a Bayesian model to investigate the incentives for the agent with the principal only knowing the mean of the output distribution. A growing literature focuses on the worst-case criteria. Edmans and Gabaix (2011)

and Carroll and Meng (2016) analyze the specific models that combine effort with additive random noises with zero mean. Chassang (2013) characterizes the lower bound on the performance of linear contracts under a dynamic setting with limited liability, moral hazard, and adverse selection. Garrett (2014) studies optimal procurement contracts with uncertain agent's preferences. In the same spirit, Antic (2014) and Carroll (2015) show the robustness and optimality of debt contracts and linear contracts when the principal is uncertain about the agent's technology and action set, respectively. Our work differs from this stream of literature in four aspects. First, our principal (the firm) is uncertain about the effort-contingent output distribution. Second, we consider robust incentive compatibility, which is absent in this literature. Third, we also characterize the optimal robust contract when the agent is risk averse. Finally, we adopt the robust optimization perspective.

Our work is also related to the literature on the max-min expected utility decision rule. This decision rule is compatible with decision makers' preference facing nonunique beliefs, which was first axiomatized by Gilboa and Schmeidler (1989) and has been commonly adopted to address uncertainty aversion in various areas, including macroeconomics (see, e.g., Hansen and Sargent 2008), finance (see, e.g., Maenhout 2004, Garlappi et al. 2007), mechanism designs (see, e.g., Bergemann and Morris 2005, Bergemann and Schlag 2011), and recently in agency theory (see, e.g., Garrett 2014, Carroll 2015). In particular, Hansen and Sargent (2008) generalize the robust control theory to investigate the issue of model uncertainty in macroeconomics. In contrast, our model is based on the robust optimization theory (Ben-Tal et al. 2009), which models uncertain parameters with known uncertainty sets to attain tractable solutions, for example, polytopes and ellipsoids. Such an approach has connections to various theories, such as stochastic programming (see, e.g., Chen et al. 2007), distributional robust optimization (see, e.g., Delage and Ye 2010), and risk theory (see, e.g., Bertsimas and Brown 2009), and has extensive applications in other areas (see Gabrel et al. 2014 for a comprehensive review). To the best of our knowledge, our work is among the first to adopt robust optimization to investigate optimal robust contract designs in agency theory.

3. Model and Formulation

We consider that a firm (i.e., the principal) hires an agent to generate an effort-contingent output for one period, for example, the sales for a sales agent. For convenience, we use "the firm" and "the principal" interchangeably. The agent can exert high or low effort, denoted by e_h and e_l , respectively, to enhance the random output Y with $e_h > e_l$. Let $c(e)$ be the cost of

effort e ($e = e_h, e_l$) with $c(e_h) > c(e_l)$. The firm observes neither the agent's effort nor the agent's realized effort cost and thus encounters a typical moral hazard problem. If the firm does not provide effort incentives, the agent will exert low effort. To incentivize the agent, the firm offers an output-contingent contract $w(\cdot)$.

We consider the game with the following sequence of moves:

1. The firm offers a contract that specifies the compensation contingent on the output realization. If the agent rejects the contract, the game terminates. If the agent accepts the contract, the game proceeds.

2. The agent chooses the effort $e \in \{e_h, e_l\}$.

3. At the end of the period, the random output Y is realized and observed by both parties. The firm makes the profit $Y - w(Y)$ and the agent receives compensation $w(Y)$.

We assume the effort-contingent random output Y follows a discrete probability distribution. Given the agent's effort e , the conditional distribution of the output is as follows:

$$Y = \begin{cases} y_n, & \text{w.p. } a_n e \\ y_{n-1}, & \text{w.p. } a_{n-1} e \\ \vdots & \\ y_1, & \text{w.p. } a_1 e \\ y_0, & \text{w.p. } 1 - \sum_{i=1}^n a_i e, \end{cases} \quad (1)$$

where $e \in \{e_h, e_l\}$ and $y_0 < y_1 < \dots < y_{n-1} < y_n$ (w.p. stands for "with probability"). The parameters $a_1, \dots, a_n$ can be interpreted as the agent's ability to enhance the output by exerting effort e . Note that the discrete output distributions are extensively adopted in the agency and salesforce literature (see, e.g., Grossman and Hart 1983, Fudenberg and Tirole 1990, Felli and Villas-Boas 2000, Dai and Jerath 2013).

Remark 1. The linear effort probability model in (1) is adopted for the sake of presentational simplicity. See Section 5.1 for the treatment of general effort-contingent probability distributions. Note that when n is sufficiently large, the class of discrete distributions can approximate a continuous distribution with arbitrary accuracy.

In contrast to the existing literature, by assuming the complete distributional information to both parties, we assume that the firm has some ambiguity or distrust over the output distribution. Specifically, we assume that the firm does not exactly know the parameters of the output distribution, that is, $a_1, \dots, a_n$, but only knows that they are in some range. This happens, for instance, when firms use statistical techniques to estimate the future output based on historical data. The confidence regions obtained for the parameter estimates, to some extent, capture their distrust on the statistical models. Such parameter uncertainty or

model uncertainty is ubiquitous in practice. Hansen and Sargent (2008) point out that decision makers, such as firms, usually have ambiguity over their statistical models. The parameter uncertainty becomes more severe when the data are not available (e.g., for new products or new agents) or are incomplete (e.g., censored data).

We model the parameter uncertainty using an uncertainty set from robust optimization theory (Ben-Tal et al. 2009). Define $x = (x_1, \dots, x_n)$, and $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$, that is, the standard norm of x . We assume that the firm knows that the parameter vector $a = (a_1, \dots, a_n)$ can be any vector in the following ellipsoid:

$$\mathbb{U} = \{(\bar{a}_1, \dots, \bar{a}_n) + (\delta_1 u_1, \dots, \delta_n u_n) \mid \|u\| \leq 1\}, \quad (2)$$

where $\delta = (\delta_1, \dots, \delta_n)$ is a nonnegative vector chosen such that $a > 0$ and $1 - \sum_{i=1}^n a_i e > 0$ for all $a \in \mathbb{U}$ (i.e., the probability of each output outcome should be positive).

To ensure all the possible distributions are well defined, we impose the following assumption.

Assumption 1. We have $\bar{a}_i - \delta_i > 0, i = 1, \dots, n$, and $\sum_{i=1}^n \delta_i^2 < (\sum_{i=1}^n \bar{a}_i - \frac{1}{e_n})^2$.

Note that $\bar{a}$ is the center of the ellipsoid and referred to as the nominal vector. If there is no parameter uncertainty, then $a = \bar{a}$ or $\delta = 0$. The vector δ measures the magnitude of the parameter uncertainty, that is, if δ_i ($i = 1, \dots, n$) is larger, then the uncertainty for a is larger. The random vector u can be viewed as random noises with zero mean where the inequality $\|u\| \leq 1$ controls the total variability of the noises. Note that we allow the arbitrary dependence among the random noise u_i .

Remark 2. To better understand the ellipsoidal uncertainty set, we consider multidimensional normal noises. For a multidimensional normal random vector with mean μ and covariance matrix Q , let $\chi_p^2(\alpha)$ be the upper 100α percentile of a χ^2 distribution with degree of freedom p ; then with probability $1 - \alpha$, the values of the random vector is inside the ellipsoid $\sqrt{(x - \mu)^T Q^{-1} (x - \mu)} \leq \sqrt{\chi_p^2(\alpha)}$. Essentially, we have a confidence level for the values in the ellipsoid. For more detailed discussions of the ellipsoidal uncertainty set, we refer readers to chapter 2 in Ben-Tal et al. (2009).

Remark 3. From a data-driven perspective, Bertsimas and Brown (2009) illustrate the inherent linkage between the norm-formed uncertainty set and the coherent risk measure. They show that all the coherent (convex) risk measures can be represented as the worst-case expected value over a family of generating distributions, that is, a convex uncertainty set is a consequence of a risk measure chosen by the decision

maker. Conversely, a convex uncertainty set also induces a coherent risk measure. For instance, polyhedral uncertainty sets correspond to Conditional Value at Risk (CVaR) measures, ellipsoidal uncertainty sets correspond to mean-variance measures, and norm-bounded uncertainty sets correspond to risk measures associated with high-order moments. In addition, if the uncertainty sets are centrally symmetric, the norm-formed uncertainty sets can be induced by the so-called distortion risk measures.

We assume that the firm is financially risk neutral. Notice that given the firm's knowledge, $\mathbb{U}$, the firm actually has infinitely many beliefs or models on the output distribution, thus it is difficult to design an effective contract. To tackle this issue, we assume a simple decision rule for the firm to evaluate contracts with parameter uncertainty: the worst-case criterion; that is, given the firm's knowledge $\mathbb{U}$, the firm maximizes its worst-case expected utility. The max-min expected utility decision rule implies that the firm is averse to parameter uncertainty or model uncertainty. However, it is compatible with a decision maker's preference facing nonunique priors and justified by the axiomatic framework of Gilboa and Schmeidler (1989). Note that this decision rule has been commonly adopted in the existing literature to address uncertainty aversion in various areas, including macroeconomics (see, e.g., Hansen and Sargent 2008), finance (see, e.g., Maenhout 2004, Garlappi et al. 2007), mechanism designs (see, e.g., Bergemann and Morris 2005, Bergemann and Schlag 2011), and recently agency theory (see, e.g., Garrett 2014, Carroll 2015).

We assume that the agent is financially risk neutral and has limited liability, that is, the payment for the agent $w(Y)$ should be no less than the agent's cost $c(e)$ for any output realization $Y = y_0, y_1, \dots, y_n$. The limited liability is commonly adopted in the agency and salesforce compensation literature (see, e.g., Park 1995; Chu and Lai 2013; Dai and Jerath 2013, 2016; Carroll 2015). In contrast to their settings, as the firm is uncertain over the underlying output distribution, we assume that any contract is also judged by its worst-case scenario of the agent. This can be explained as follows: regardless of whether the agent knows the exact values of a , from the firm's perspective, if it would like to incentivize the agent to exert effort e , the firm should ensure that the agent has incentive to exert effort e for any possible $a \in \mathbb{U}$ based on its knowledge. In this sense, the agent is also averse to the parameter uncertainty.

Define

$$\Delta c = \frac{c(e_h) - c(e_l)}{e_h - e_l},$$

which is the marginal effort cost. We assume that Δc is small enough, or, alternatively, $y_1, \dots, y_n$ are sufficiently large such that the firm prefers inducing high

effort and a feasible contract does exist. Note that a fixed payment $w(Y) \equiv c(e_l)$ is sufficient to induce low effort. We next focus on how to induce high effort from the agent. For ease of exposition, define $w_i \equiv w(y_i)$ for $i = 0, \dots, n$. Then, the firm's contract design problem to induce high effort can be formulated as follows:

$$\begin{aligned} & \pi(e_h) \\ &= \max_{w_i, i=0,1,\dots,n} \min_{a \in \mathbb{U}} \left[\sum_{i=1}^n a_i e_h (y_i - w_i) + (y_0 - w_0) \left(1 - \sum_{i=1}^n a_i e_h \right) \right], \end{aligned} \quad (3)$$

$$\begin{aligned} & \text{subject to } \sum_{i=1}^n a_i e_h w_i + w_0 \left(1 - \sum_{i=1}^n a_i e_h \right) - c(e_h), \\ & \geq \sum_{i=1}^n a_i e_l w_i + w_0 \left(1 - \sum_{i=1}^n a_i e_l \right) \\ & \quad - c(e_l), \forall a \in \mathbb{U}, \quad (4) \\ & w_i \geq c(e_h), \forall i = 0, 1, \dots, n. \quad (5) \end{aligned}$$

We refer to this problem as the *robust contract design problem*, any feasible solution as the *robust contract*, and the optimal solution as the *optimal robust contract*. As the firm does not know the exact values of a and assumes that $a \in \mathbb{U}$, the incentive compatible condition should hold for any $a \in \mathbb{U}$. We refer to (4) as the *robust incentive compatibility*. The objective in (3) gives the worst-case performance under optimal contracting for the firm to incentivize the agent to exert high effort. Note that the robust contract design problem in (3)–(5) has infinitely many constraints for the robust incentive compatibility. It is unclear which contract form is robust to the parameter uncertainty.

Remark 4. Note that the limited liability is not crucial for our results. In fact, our results are robust to the case with the (robust) participation constraint, that is, the worst-case expected utility for exerting high effort is no less than an outside option, that is, $\mathbb{E}_{Y|e_h}[w(Y)] \geq c(e_h) + V_0$ for all $a \in \mathbb{U}$, where V_0 is the valuation of the outside option. We refer readers to Online Appendix C for a detailed discussion.

In particular, when $\delta_i = \delta$, for all $i = 1, \dots, n$, based on standard statistics, as δ becomes larger, the firm has more confidence in its belief but its worst-case performance is worse off. For example, suppose that the confidence level of the firm is $\Pr(\mathbb{U}) = q(\delta)$ for some function $q(\cdot)$. If the firm has a choice, the interesting research question is how the firm should choose δ to trade off between its worst-case performance and confidence level.

4. Analysis and Results

In this section, we first present a benchmark case without parameter uncertainty. Then, in the presence

of the parameter uncertainty, we show that to incentivize the risk-neutral agent to exert high effort, a simple linear (commission-based) contract in the form of $c(e_h) - \alpha y_0 + \alpha Y$, for some $\alpha \in [0, 1]$ and $Y = y_1, \dots, y_n$, is optimal and robust to the parameter uncertainty. No other contracts are optimal. Finally, we show that when the agent is financially risk averse and has a piecewise linear utility, the only optimal robust contract is a piecewise linear contract that consists of progressive fixed payments and linear rewards with progressive commission rates.

4.1. Benchmark Case Without Parameter Uncertainty

To provide a benchmark, we consider the optimal contract without parameter uncertainty, that is, $\delta = (0, \dots, 0)$.

Lemma 1. Without parameter uncertainty, (i) to induce low effort e_l , it is optimal for the firm to offer a fixed payment contract, that is, $w^*(Y) \equiv c(e_l)$; and (ii) to induce high effort e_h , it is optimal for the firm to set $w_0^* = c(e_h)$ and any $(w_1^*, \dots, w_n^*)$ such that $\sum_{i=1}^n \bar{a}_i (w_i^* - w_0^*) = \Delta c$ and $w_i^* \geq w_0^*$ for $i = 1, \dots, n$.

These results are intuitive and standard in the existing literature. For part (i), if the firm does not want to induce high effort, it can simply offer a fixed payment. One can verify that this is also true in the presence of parameter uncertainty. For part (ii), when the firm would like to induce high effort, a higher-powered contract rather than a fixed payment contract is needed. We can verify that there are in fact infinitely many optimal contracts including a linear one.

Remark 5. When the distribution satisfies the strict MLRP, only a quota-bonus contract is optimal without parameter uncertainty (see, e.g., Oyer 2000; Chu and Lai 2013; Dai and Jerath 2013, 2016).

4.2. Robust Contract

We next analyze the firm's optimal robust contract design problem with parameter uncertainty when the agent is financially risk neutral. We refer to this problem as the *basic model*. The program in (3)–(5) can be written as follows:

$$\begin{aligned} & \pi(e_h) \\ &= \max_{w_i, i=0,1,\dots,n} \min_{a \in \mathbb{U}} \sum_{i=1}^n a_i e_h [y_i - y_0 - (w_i - w_0)] + y_0 - w_0, \end{aligned} \quad (6)$$

$$\text{subject to } \sum_{i=1}^n a_i (w_i - w_0) \geq \Delta c, \forall a \in \mathbb{U}, \quad (7)$$

$$w_i \geq c(e_h), \forall i = 0, 1, \dots, n. \quad (8)$$

Notice that under the optimal robust contract, we must have $w_0^* = c(e_h)$; otherwise, we can derive a better

contract for the firm by decreasing w_0 slightly without violating any constraints. For notional simplicity, we define $x_i \equiv w_i - w_0$, which can be viewed as the bonus payment for realization y_i for $i = 1, \dots, n$. Then, the limited liability is equivalent to $x \geq \mathbf{0}$. We then simplify the program in (6)–(8) as follows:

$$\pi(e_h) = \max_{x \geq \mathbf{0}} \min_{a \in \mathbb{U}} \sum_{i=1}^n a_i e_h [y_i - y_0 - x_i] + y_0 - w_0, \quad (9)$$

$$\text{subject to } \sum_{i=1}^n a_i x_i \geq \Delta c, \forall a \in \mathbb{U}. \quad (10)$$

Notice that the robust incentive compatibility (10) has infinitely many incentive-compatible constraints. We utilize the recent developments in robust optimization to simplify the program in (9)–(10) in the following lemma.

Lemma 2 (Ben-Tal et al. 2009). *Consider the following minimization problem:*

$$U(x) = \min_{a \in \mathbb{U}} \langle x, a \rangle, \quad (11)$$

where $\langle \cdot \rangle$ denotes the inner product. Then, $U(x) = \sum_{i=1}^n a_i x_i - \sqrt{\sum_{i=1}^n \delta_i^2 x_i^2}$. In particular, $U(x)$ is concave in x .

Lemma 2 explicitly characterizes the worst-case performance $U(x)$ under the ellipsoidal uncertainty set $\mathbb{U}$, which includes a nominal performance $\sum_{i=1}^n \bar{a}_i x_i$ and a norm-form uncertainty-adjusted term $\sqrt{\sum_{i=1}^n \delta_i^2 x_i^2}$ due to the uncertainty aversion to the parameter uncertainty or model uncertainty. The economic implication from Lemma 2 is that the principal with the max-min preference subject to the ellipsoidal parameter uncertainty set is equivalent to a principal with the utility function $U(x)$. We therefore refer to $U(x)$ as the surrogate utility. One may notice that $U(x)$ resembles the mean-variance utility. Indeed, our results apply to the setting where the agent has a mean-variance utility (see Section 5.4).

Lemma 3. *The robust contract design problem in (9) and (10) is equivalent to the following optimization problem:*

$$\max_{x \geq \mathbf{0}} V(y, x) \quad (12)$$

$$\text{subject to } U(x) \geq \Delta c, \quad (13)$$

where

$$V(y, x) = \sum_{i=1}^n \bar{a}_i (y_i - y_0 - x_i) - \sqrt{\sum_{i=1}^n \delta_i^2 (y_i - y_0 - x_i)^2}.$$

With Lemma 3, we transform the infinitely many incentive compatible constraints into one simple convex constraint and the robust contract design problem into a simple convex optimization problem. There are several interesting observations on the program in (12) and (13). First, notice that both $V(y, x)$ and $U(x)$ are concave in x . It is equivalent to a risk-averse firm with surrogate

utility $V(y, x)$ contracting with a risk-averse agent with surrogate utility $U(x)$. Second, the robust incentive compatibility reduces to a form of the participation constraint (13), which must be binding under the optimal contract. Notice that when $x_i \geq 0$, the agent's surrogate utility $U(x)$ is strictly increasing in x_i while $V(y, x)$ is decreasing in x_i for $i = 1, \dots, n$. Given these properties, it is clear that under the optimal solution, the constraint in (13) must be binding. This indicates that the firm should design a contract that maximizes its own surrogate utility but only leaves the agent the reservation surrogate utility Δc . With these observations, we can transform the optimal robust contract design problem into a social optimization problem.

We consider the following social optimization problem:

$$\max_{x \geq \mathbf{0}} [V(y, x) + U(x)] \quad (14)$$

$$= \max_{x \geq \mathbf{0}} \sum_{i=1}^n \bar{a}_i (y_i - y_0) - \sqrt{\sum_{i=1}^n \delta_i^2 x_i^2} - \sqrt{\sum_{i=1}^n \delta_i^2 (y_i - y_0 - x_i)^2}, \quad (15)$$

that is, it chooses the optimal bonus payment schedule x to maximize the total surrogate utility.

Lemma 4. *Any $x = \alpha(y_1 - y_0, \dots, y_n - y_0)$ for $\alpha \in [0, 1]$ is optimal to the social optimization problem in (14). No other solutions are optimal.*

This result states that a linear solution is uniquely optimal to the social optimization problem in (14). Intuitively, based on (15), to find the optimal solution, it is equivalent to finding a point with the shortest distance between the point $\mathbf{0}$ and the point $(y_1 - y_0, \dots, y_n - y_0)$.

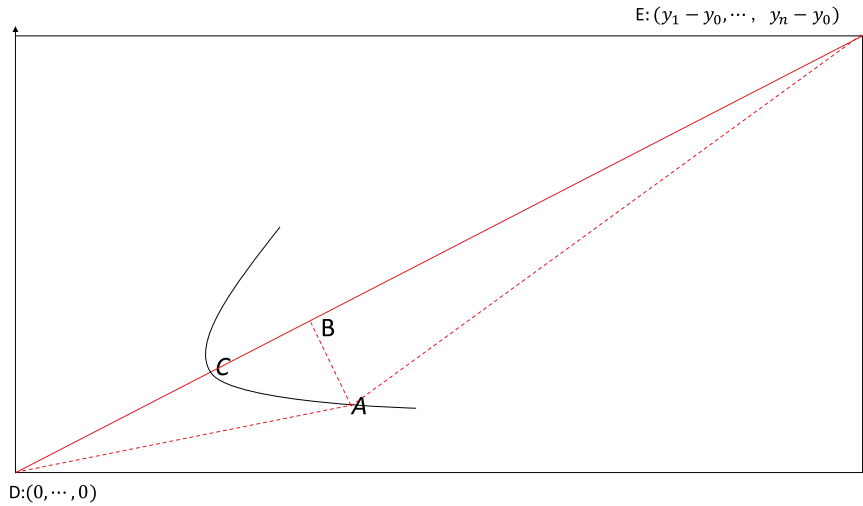
Proposition 1. *The optimal robust contract is a linear contract, $w^*(Y) = \alpha^* Y - \alpha^* y_0 + c(e_h)$, where the base compensation is $-\alpha^* y_0 + c(e_h)$, the bonus under output Y is $\alpha^* Y$ for all $Y = y_i$ for $i = 1, \dots, n$, and the optimal commission rate is*

$$\alpha^* = \frac{\Delta c}{\sum_{i=1}^n \bar{a}_i (y_i - y_0) - \sqrt{\sum_{i=1}^n \delta_i^2 (y_i - y_0)^2}}. \quad (16)$$

Moreover, any other type of contract is not optimal. In particular, the optimal commission rate α^* is increasing in δ_i and decreasing in y_i for all $i = 1, \dots, n$.

Remark 6. More rigorously, our contract form in Proposition 1 should be referred to as an affine contract instead of a linear contract.

To obtain the main idea of the proof, we illustrate the optimality of a linear contract in Figure 1. In Figure 1, on the hyperplane, the top right point is $E : (y_1 - y_0, \dots, y_n - y_0)$, and the bottom left point is $D : \mathbf{0}$. Line $\overline{AB}$ is perpendicular to line $\overline{DE}$. Any point x

Figure 1. (Color online) Illustration of the Optimality of a Linear Contract

Note: The distance from x to $\mathbf{0}$ is measured by $\sqrt{\sum_{i=1}^n \delta_i^2 x_i^2}$.

in the rectangle to the right of the indifference curve, which is defined by $U(x) = \Delta c$, is a feasible solution to the robust contract design problem in (12). It is clear that any point that is not on the indifference curve should not be optimal (either the robust incentive compatibility does not hold or the firm's performance can be improved).

Suppose that point A , which is on the indifference curve but not on $\overline{DE}$, is optimal. The total social welfare defined in (15) is a constant $\sum_{i=1}^n \bar{a}_i(y_i - y_0)$ minus the total distance or norm of $\overline{AD}$ and $\overline{AE}$ (the distance from x to $\mathbf{0}$ is $\sqrt{\sum_{i=1}^n \delta_i^2 x_i^2}$). We shall argue that A cannot be optimal by contradiction. Indeed, if we move A to B on $\overline{DE}$, the total distance of $\overline{BD}$ and $\overline{BE}$ is strictly smaller than the counterpart of $\overline{AD}$ and $\overline{AE}$. So by moving from A to B , we obtain strictly pareto improvements for both the firm and the agent. This shows that A cannot be optimal. Then, C , which is the intersection of the indifference curve and $\overline{DE}$, must be optimal. This shows that only a linear contract is optimal.

Interestingly, the optimal robust contract consists of two parts: a base payment and a linear bonus. Supposedly, nonlinear contracts are flexible and should be superior to linear contracts. However, Proposition 1 suggests that no other contract forms, including nonlinear contracts, are robust to the parameter uncertainty. What is the intuition behind the conflicting results? Recall that with the parameter uncertainty under the worst-case criteria, both the firm and the agent's original risk-neutral utilities become risk-averse surrogate utilities that have the homogeneous uncertainty-adjusted terms of the norm-form depending on extra output $y_i - y_0$ ($i = 1, \dots, n$). To share the output

between these two parties with homogenous risk-averse surrogate utilities, a linear contract, which is a risk-sharing contract, is required. Essentially, with the parameter uncertainty, to incentivize the agent, a quota-based bonus payment is not sufficient as both the firm and the agent exhibit a risk-aversion property with the uncertainty-adjusted terms dependent on $y_i - y_0$ ($i = 1, \dots, n$).

Notice that the commission rate α^* depends on δ , in particular, α^* is increasing in δ . We can interpret this from the informativeness of δ (see Hölmstrom 1979 for the definition of informativeness). Because of the parameter uncertainty, the output realization Y alone is no longer a sufficient signal for the firm to infer the agent's effort. To design an effective contract, the firm needs more information beyond that conveyed by Y to enhance the firm's power of inference, which is provided by δ . To see this, the smaller δ is, the more accurate information the firm has. On the other hand, for larger δ , the agent demands a higher commission rate, which means the parameter uncertainty indeed benefits the agent. Hence, the agent has no incentive to provide more accurate market information to the firm even if the agent knows the true distribution. Moreover, as long as $\delta > 0$, a linear contract is uniquely optimal regardless of how small $\|\delta\|$ is. Our results thus provide a new explanation for the popularity of linear contracts or commission-based contracts in practice: the parameter uncertainty can be the driving factor.

We next analyze the impact of the parameter uncertainty on the firm's worst-case performance. The following proposition explicitly characterizes the firm's worst-case performance under the optimal robust contract.

Proposition 2. Under the optimal robust contract with effort e_h , the firm's worst-case payoff is

$$\pi(e_h) = \left[\sum_{i=1}^n \bar{a}_i(y_i - y_0) - \sqrt{\sum_{i=1}^n \delta_i^2 (y_i - y_0)^2} - \Delta c \right] e_h + y_0 - c(e_h).$$

In particular, if for all $i = 1, \dots, n$, $\delta_i = \delta$, then

$$\pi(e_h) = \left[\sum_{i=1}^n \bar{a}_i(y_i - y_0) - \delta \sqrt{\sum_{i=1}^n (y_i - y_0)^2} - \Delta c \right] e_h + y_0 - c(e_h),$$

and the worst-case performance linearly decreases in δ at rate $e_h \sqrt{\sum_{i=1}^n (y_i - y_0)^2}$.

It is rather intuitive to see that the parameter uncertainty has an adverse impact on the firm's worst-case performance: the more ambiguous over the parameters the firm is, the more the firm has to pay the agent and the less profit it can make given the same output. In particular, if for all $\delta_i = \delta (i = 1, \dots, n)$, the firm's worst-case performance decreases linearly in δ . We can interpret δ as the confidence level of $\mathbb{U}$ (as δ increases, the confidence level becomes higher). This implies that the firm should carefully choose its confidence level as there is a price for robustness.

We next analyze the trade-off between the confidence level of $\mathbb{U}$ and the firm's worst-case performance. Suppose that the value of confidence level is $K \Pr(\mathbb{U}) = Kq(\delta)$, which is increasing in δ , for some constant K . Then the optimal choice of δ is defined as

$$\delta^* = \arg \max_{\delta \geq 0} \left[Kq(\delta) - \delta e_h \sqrt{\sum_{i=1}^n (y_i - y_0)^2} \right]. \quad (17)$$

Proposition 3. The optimal choice δ^* is decreasing in e_h and y_i for all $i = 1, \dots, n$.

Proposition 3 indicates that the firm must strike a balance between its confidence level and its worst-case performance guarantee. When $y_i, i = 1, \dots, n$ is higher, the firm should choose a smaller δ , even if it is less confident about the corresponding $\mathbb{U}$. Because of the linearity of the optimal robust contract, a higher y_i means a higher price of robustness for the firm. To compensate for the high information cost, it is better for the firm to lower its confidence level and obtain more accurate information. On the other hand, a larger e_h strengthens the agent's role in generating higher outputs. Therefore, the agent would demand a higher commission rate for the same output, which again leads to high information cost and a reduction of the firm's confidence level.

Remark 7. Note that Carroll (2015) shows that the optimal worst-case contract is a linear contract in the form αY ($\alpha \in [0, 1]$) under action set uncertainty, that is, the principal is uncertain about what the agent can and cannot do. The author's contract form differs from ours as our contract has a base compensation component in the form of $c(e_h) - \alpha y_0$. Clearly, his setting is also different from our setting. We focus on the parameter uncertainty instead of the action set uncertainty. Carroll (2015) does not consider the robust incentive compatibility as he considers the setting where the agent knows more about his/her own action set or technology choice than the principal.

4.3. Robust Contract with the Risk-Averse Agent

In the previous section, we show that when the agent is financially risk neutral, the simple linear contract is uniquely optimal in the presence of the parameter uncertainty. One may ask what if the agent is risk averse. In this section, we show that when the risk-averse agent has a piecewise linear concave utility, the only optimal robust contract is a piecewise linear contract that consists of progressive fixed payments and linear rewards with progressive commission rates.

Specifically, we assume the agent has the following utility function:

$$u(x) = \min_{s \in \{1, 2, \dots, S\}} \{k_s + b_s x\}, \quad (18)$$

where $0 = k_1 < \dots < k_S, b_1 > \dots > b_S > 0$. Let $\mathcal{Q} = \{\bar{w}_1, \dots, \bar{w}_{S-1}\}$ be the set of all changeover points of the utility function (see Figure 2 for an illustration of the utility function).

Remark 8. Note that any concave function can be approximated by piecewise linear concave functions with arbitrary accuracy. In this sense, the optimal contract derived from a piecewise linear utility function can be arbitrarily close to the optimal robust contract associated with any type of risk-averse utilities.

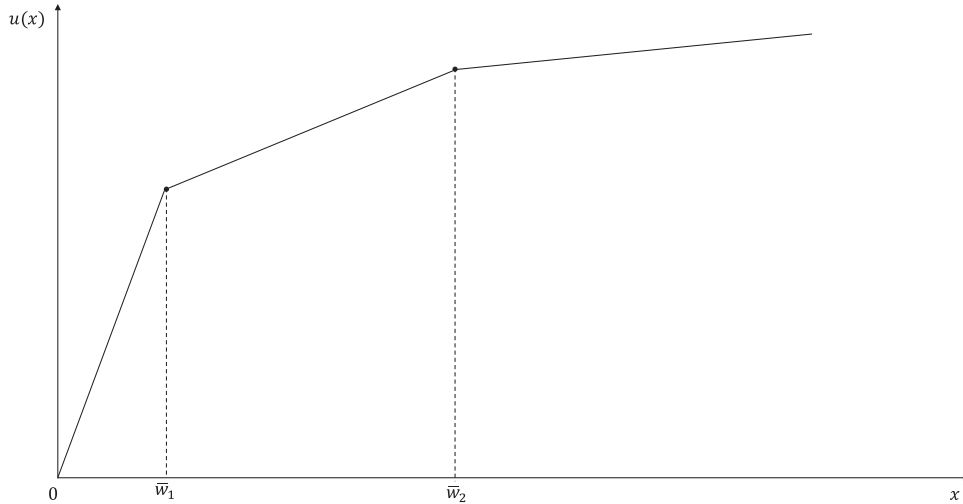
To incentivize the agent with the piecewise linear concave utility, the firm's optimal robust contract design problem is provided as follows:

$$\max_{w_i, i=0, \dots, n} \min_{a \in \mathbb{U}} \sum_{i=1}^n a_i e_h [y_i - y_0 - (w_i - w_0)] + y_0 - w_0, \quad (19)$$

$$\text{subject to } \sum_{i=1}^n [u(w_i) - u(w_0)] a_i \geq \Delta c, \forall a \in \mathbb{U}, \quad (20)$$

$$u(w_i) \geq c(e_h), i = 0, \dots, n. \quad (21)$$

Without loss of generality, we normalize $y_0 = 0$, and we reduce the limited liability to $u(w_i) \geq 0$ for all $i = 0, \dots, n$. Note that the limited liability only changes

Figure 2. Piecewise Linear Concave Utility of Changeover Points $\bar{w}_1, \bar{w}_2$ 

the base payment and does not affect the optimal robust contract structurally (our results are robust to the participation constraint; see Online Appendix C). Then, based on Lemma 2, we can transform this problem to an equivalent convex optimization problem with only one robust incentive compatible constraint.

Lemma 5. *It is optimal to have $w_0^* = u^{-1}(0) = 0$. Under Assumption 1, problem (19)–(21) is equivalent to the following convex optimization problem:*

$$\max_{w \geq 0} V(y, w), \quad (22)$$

$$\text{subject to } U(w) \geq \Delta c, \quad (23)$$

where

$$w = (w_1, \dots, w_n),$$

$$V(y, w) = \sum_{i=1}^n \bar{a}_i (y_i - w_i) - \sqrt{\sum_{i=1}^n \delta_i^2 (y_i - w_i)^2},$$

$$U(w) = \sum_{i=1}^n \bar{a}_i u(w_i) - \sqrt{\sum_{i=1}^n \delta_i^2 u(w_i)^2}.$$

Moreover, under the optimal contract, the constraint (23) must be binding.

Note that the agent utility function is not continuously differentiable. However, we can utilize the subgradient of a concave function to obtain the first-order optimality condition. Let ∂u be the set of subgradients of u . It is clear that $\partial u(x) = b_{s-1}$ for $x \in (\bar{w}_{s-1}, \bar{w}_s)$ and $\partial u(\bar{w}_s) = [b_s, b_{s-1}]$ for $s = 1, \dots, S$, where $\bar{w}_0 \equiv 0$ and $\bar{w}_S = \infty$. Define the Lagrangian function as $\mathcal{L}(w, \lambda) = V(y, w) + \lambda[U(w) - \Delta c]$; then, the optimality condition (Karush-Kuhn-Tucker condition; see

theorem 27.4 of Rockafellar 1972) for the program in (22)–(23) is

$$\begin{cases} -\bar{a}_i + \frac{\delta_i^2 (y_i - w_i)}{\sqrt{\sum_{l=1}^n \delta_l^2 (y_l - w_l)^2}} + \lambda \partial u(w_i) \left[\bar{a}_i - \frac{\delta_i^2 u(w_i)}{\sqrt{\sum_{l=1}^n \delta_l^2 u(w_l)^2}} \right] \ni 0, \\ i = 1, \dots, n \\ \lambda [U(w) - \Delta c] = 0 \\ \lambda \geq 0. \end{cases} \quad (24)$$

Note that $U(w) = \Delta c$ at the optimum. Then we have $\lambda > 0$, and for the time being we ignore the limited liability that can be verified ex post. Let $w^*(y)$ denote the minimum optimal compensation under the output y . Notice that the problem is similar to the basic model except the optimality conditions for the changeover points of u , that is, the points in the set $\mathcal{Q}$.

Lemma 6. *Under the optimal robust contract $w^*(y)$, we have the following results: (i) $w^*(y)$ is nondecreasing in y ; (ii) if $\bar{w}_s \in \mathcal{Q}$ ($s \in \{1, \dots, S-1\}$) is attainable under $w^*(y)$, then there exists some interval, $\mathcal{I}_s = [\underline{y}_s, \bar{y}_s]$, such that $w^*(y) \equiv \bar{w}_s$ for all $y \in \mathcal{I}_s$; and (iii) if $\bar{w}_{s+1}$ is also attainable under $w^*(y)$, then $\bar{y}_s < \underline{y}_{s+1}$, that is, $\mathcal{I}_s$ is increasing and disjoint with $\mathcal{I}_{s+1}$.*

Note that the monotonicity of $w^*(y)$ in part (i) is an appealing property to avoid gaming behavior of the agent (Innes 1990). It directly follows from that $V(y, w)$ is supermodular in (w_i, y_i) . Part (ii) indicates that the optimal robust contract consists of progressive flat payments to the output at the changeover points $\mathcal{Q}$, that is, fixed payments $\bar{w}_s \in \mathcal{Q}$. The underlying intuition is that $\frac{\partial \mathcal{L}}{\partial w_i}$ is decreasing in w_i and jumps downward whenever w_i goes across any changeover point $\bar{w}_s \in \mathcal{Q}$.

from its left to its right. This implies that $\bar{w}_s$ is a local optimum thus the global optimum can be $w_i^* = w^*(y_i)$ for some interval $\mathcal{J}_s$. Part (iii) is clear given the monotonicity of the optimal robust contract and the concavity of the utility function.

We next characterize the optimal robust contract that does not attain the changeover points, that is, for $w^*(y) \notin \mathcal{Q}$.

Lemma 7. Under the optimal robust contract w^* , we have the following results:

- The optimal contract w^* must solve the social optimization problem $\max_{w \geq 0} [V(y, w) + U(w)]$.
- Under the optimal contract, we must have $w^*(y) = \frac{\beta y - b_j k_j}{\beta + (b_j)^2}$ if $w^*(y) \in (\bar{w}_{j-1}, \bar{w}_j)$ for some $j \in \{1, \dots, S\}$, where $\beta > 0$ is chosen such that $U(w^*) = \Delta c$.

Interestingly, Lemma 7 indicates that except for those fixed payments at $\mathcal{Q}$, the optimal robust contract $w^*(y)$ is linear in y for any $w^*(y) \in (\bar{w}_{j-1}, \bar{w}_j)$. The linearity here is again due to its robustness to the parameter uncertainty. In this sense, Lemma 7 generalizes the linear contract in the basic model to the piecewise linear concave utility. Furthermore, notice that $\frac{\beta}{\beta + (b_j)^2}$ is increasing in b_j given β . It implies that the optimal commission rate is increasing in the output. This is intuitive as the agent has a less marginal utility to higher compensation, a higher commission rate is required.

It worth noting that the optimal robust contract may not attain all changeover points $\mathcal{Q}$. Suppose there are $T \in \{1, \dots, S-1\}$ changeover points contained under the optimal robust contract, then we can construct a partition of $[0, y_n]$ based on $\mathcal{J}_1, \dots, \mathcal{J}_T$. Define $\mathcal{Y}_1 = [0, y_1]$, $\mathcal{Y}_j = (\bar{y}_j, y_{j+1})$ for $j = 2, \dots, T$, and $\mathcal{Y}_{T+1} = (\bar{y}_T, y_n]$ if $\bar{y}_T < y_n$, or $\emptyset$ otherwise. The following result follows directly from Lemmas 6 and 7.

Proposition 4. The optimal robust contract can be described as follows:

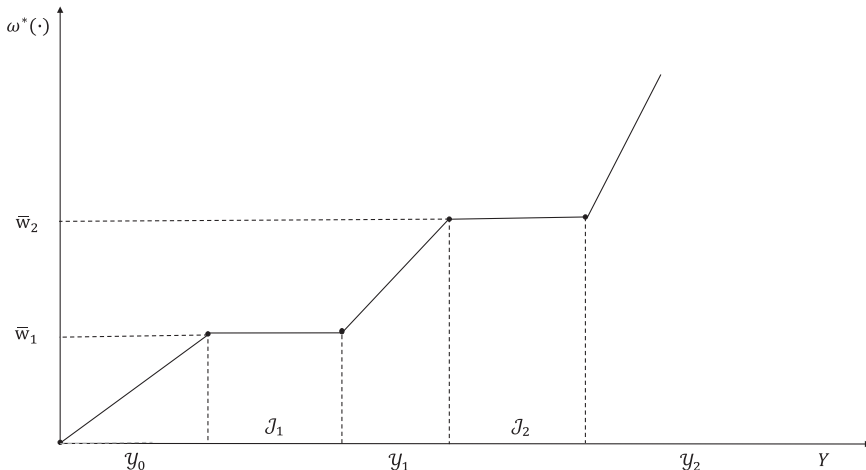
$$w^*(y) = \begin{cases} \bar{w}_s & \text{for } y \in \mathcal{J}_s, s \in \{1, \dots, T\} \\ \bar{w}_{j-1} + \alpha_j^*(y - \bar{y}_{j-1}) & \text{for } y \in \mathcal{Y}_j, j \in \{1, \dots, T+1\} \end{cases}$$

where T is the number of quotas such that $T \leq S-1$, $\alpha_j^* \in (0, 1)$ is increasing in j , and $\mathcal{J}_s, \mathcal{Y}_j$ are defined as previously.

Proposition 4 states that the optimal robust contract with a piecewise linear concave agent utility only consists of progressive fixed payments in $\mathcal{Q}$ and linear rewards with progressive commission rates. Figure 3 illustrates the structure of the optimal robust contract. We refer to this type of contract as the piecewise linear contract.

One may argue that the fixed compensations capping the reward for certain performance intervals and multiple thresholds (quotas) in this piecewise linear contract seem uncommon at the first glance, for example, $\mathcal{J}_1, \mathcal{J}_2$ in Figure 3. However, it turns out that such performance caps and multiple thresholds in incentive contracts are in fact quite common in practice. Based on 177 public companies in the United States, Murphy (2000) reports that 80% of companies use capped bonuses and multiple performance thresholds in their CEO compensation. In particular, a typical compensation plan pays nothing below some threshold, then linear incentives that are capped after another higher threshold (see figure 1 in Murphy 2000). Interestingly, our results are consistent with the previous typical compensation plans. Hence, our results provide theoretical justification of the commonly used piecewise linear contracts: the robustness of linearity may account for their popularity, whereas the performance caps and multiple thresholds provide flexibility for the risk sharing between the principal and the agent under moral hazard.

Figure 3. Illustration of the Optimal Robust Contract with Piecewise Linear Concave Utility



On the other hand, Basu et al. (1985) argue that the optimal contracts can be best approximated by the piecewise linear contracts for the ease of implementation, and that many popular contracts, including commission- or quota-based contracts, can be viewed as special cases of piecewise linear contracts. Proposition 4 echoes their argument by showing that the piecewise linear contract may indeed to be uniquely optimal. Furthermore, Proposition 4 suggests firms should set performance caps and multiple performance quotas to augment their compensation schemes. As another sharp contrast, Basu et al. (1985) show that the optimal contract can be convex or concave depending on the agent's risk-aversion characteristics under some specific output distributions and utility functions. In contrast, our contract is not necessarily convex in the output. It is clear from Figure 3 that the optimal robust contract is generally neither concave nor convex due to the existence of the changeover points $\mathcal{Q}$ of the utility function.

Remark 9. We note that the structure of piecewise linear contracts holds for any utility with finitely many pieces or finitely many changeover points. Consequently, the optimal robust contract should consist of finitely many pieces and progressive commission rates, per Proposition 4. Although smooth utilities are also common in the literature, we argue that the piecewise linear utilities are sufficient to provide good approximations and can yield effective incentive contracts; recall that typical CEO compensation plans only have two pieces (Murphy 2000).

5. Robustness Checks

In this section, we show that our results are robust to a variety of settings: (1) general output distributions, (2) general ℓ^p -norm uncertainty sets, (3) multiple effort levels, and (4) mean-variance risk-averse agent utility. In summary, parameter uncertainty or model uncertainty can be viewed as one of the driving factors for the robustness, or popularity, of linear contracts. For brevity, we relegate the detailed analyses and proofs to Online Appendix B. Because robustness of the corresponding piecewise linear contracts can be treated analogously, it is omitted. We discuss how to incorporate the participation constraint and inconsistent information sets in Online Appendix C.

5.1. General Output Distribution

In the basic model, we assume a linear effort-output relationship, that is, $\Pr(Y = y_i|e) = a_i e$ for $i = 1, \dots, n$. We next generalize our results to a general effort-

contingent output distribution with model uncertainty, that is, $\Pr(Y = y_i|e) = p_i(e)$, $i = 1, \dots, n$. In this sense, our results are robust to model uncertainty with a general output distribution.

Specifically, we assume the conditional output distribution given the agent's effort $e \in \{e_h, e_l\}$ is as follows:

$$Y = \begin{cases} y_n, & \text{w.p. } p_n(e) \\ y_{n-1}, & \text{w.p. } p_{n-1}(e) \\ \vdots & \\ y_1, & \text{w.p. } p_1(e) \\ y_0, & \text{w.p. } 1 - \sum_{i=1}^n p_i(e). \end{cases} \quad (25)$$

Instead of assuming uncertain parameters, we assume that the firm has ambiguity on the probability mass functional $\mathbf{p}(e) = (p_1(e), \dots, p_n(e))$ and only know that $\mathbf{p}(e)$ can be any vector in the following uncertainty set:

$$\mathbb{U}(e) = \{(\bar{p}_1(e), \dots, \bar{p}_n(e)) + f(e)(\delta_1 \xi_1, \dots, \delta_n \xi_n) \mid \|\xi\| \leq 1\}, \quad (26)$$

where $\bar{\mathbf{p}}(e)$ is the nominal functional of $\mathbf{p}(e)$, δ is the vector of nonnegative constants representing the firm's confidence level for the conditional probability distribution, $f(e)$ denotes the impact of the agent's effort e on the uncertainty set, and ξ are random noises whose total variability is bounded. Therefore, δ and $f(e)$ jointly capture the magnitude of the model uncertainty. As $|f(e)|$ or δ_i ($i = 1, \dots, n$) increases, the uncertainty set becomes larger. We further impose the following assumption on the uncertainty set.

Assumption 2. For all $i = 1, \dots, n$ and $e \in \{e_h, e_l\}$, (i) $\bar{p}_i(e) > \delta_i |f(e)|$, and $|f(e)| \sqrt{\sum_{i=1}^n \delta_i^2} < 1 - \sum_{i=1}^n \bar{p}_i(e)$; (ii) $|\bar{p}_i(e_h) - \bar{p}_i(e_l)| \geq \delta_i |f(e_h) - f(e_l)|$; and (iii) $\mathbf{p}(e)$ satisfies the (weak) MLRP, that is, $\frac{1 - \sum_{i=1}^n \bar{p}_i(e_h)}{1 - \sum_{i=1}^n \bar{p}_i(e_l)} \leq \frac{\bar{p}_1(e_h)}{\bar{p}_1(e_l)} \leq \dots \leq \frac{\bar{p}_n(e_h)}{\bar{p}_n(e_l)}$.

Part (i) ensures that the conditional probability distributions are well defined, that is, the probability for each output outcome is positive. Part (ii) requires that the firm's ambiguity of each point mass not exceed the difference between the nominal conditional distributions. Part (iii) is the standard assumption in the classical agency theory, that is, the MLRP for the nominal distribution. That is, a higher performance is more informative of the agent's effort under the nominal distribution. However, as we shall see later, with model uncertainty, this property implies that a higher performance is more profitable than a lower performance under the worst-case criterion.

Then, the firm's robust contract design problem can be formulated as follows:

$$\begin{aligned}
 \pi(e_h) &= \max_{w_i, i=0, \dots, n} \min_{p(e_h) \in \mathbb{U}(e_h)} \sum_{i=1}^n [y_i - y_0 - w_i + w_0] \\
 &\quad \times p_i(e_h) + y_0 - w_0. \quad (27) \\
 \text{subject to } &\sum_{i=1}^n (w_i - w_0) p_i(e_h) - c(e_h) \\
 &\geq \sum_{i=1}^n (w_i - w_0) p_i(e_l) - c(e_l), \forall p(e) \in \mathbb{U}(e). \quad (28) \\
 &w_i \geq c(e_h), i = 0, \dots, n. \quad (29)
 \end{aligned}$$

It is clear that when $p_i(e) = a_i e$ ($i = 1, \dots, n$) and $f(e) = e$ for $e = e_h, e_l$, this problem regresses to the basic model. One can verify that the basic model satisfies Assumption 2, therefore, it is indeed a special case of the general output distribution. Following the same logic in the basic model, we can derive the optimal robust contract for the previous problem, provided as follows.

Proposition 5. Under Assumption 2, the optimal robust contract is the following generalized linear contract:

$$w^*(y_i) = \begin{cases} c(e_h), & \text{for } y_i \leq y_k, \\ \alpha^*(y_i - y_0) + c_i & \text{for } y_i > y_k, \end{cases} \quad (30)$$

where $k = \max\{i \mid \bar{p}_i(e_h) - \bar{p}_i(e_l) \leq 0, i = 0, 1, \dots, n\}$, c_i is a constant for $i = k, \dots, n$, and $\alpha^* \in (0, 1)$. Moreover, no other contract is optimal.

Proposition 5 states that under a general output distribution with MLRP, when facing model uncertainty, it is optimal for the firm to use commissions to reward the agent if the agent's output exceeds some quota (threshold) $\bar{y} = y_k$. The contract in (30) is a generalization of the linear contract defined in Proposition 1. We refer to this type of contract as the *generalized linear contract*. Clearly, the quota is simply y_0 under the linear contract in the basic model.

To understand the structure, note that MLRP implies the single-crossing property of the nominal distributions, that is, for some k , $\bar{p}_i(e_h) - \bar{p}_i(e_l) \leq 0$ for all $i \leq k$ and $\bar{p}_i(e_h) - \bar{p}_i(e_l) > 0$ otherwise (noting that under the basic model, that is, the linear effort-output model, $k = 1$). It further implies that for any realization $y_i \leq \bar{y} = y_k$, the marginal benefit for the agent from exerting high effort rather than low effort is negative but positive for those realizations above $\bar{y}$. Therefore, the agent only has incentives to exert high effort to achieve higher performance above $\bar{y}$. Part (ii) of Assumption 2 ensures that the single-crossing property preserves under the worst-case criterion. As a result, the firm does not need to provide incentives

for those lower outputs ($y_i \leq \bar{y}$). As for the higher outputs ($y_j > \bar{y}$), a risk-sharing scheme is required due to the presence of model uncertainty (following the same logic as in the basic model). Again, the linearity is driven by the robustness to model uncertainty. In this sense, our results are still robust to general output distributions. Finally, we note that our results apply to any discrete output distribution with finite outcomes, particularly for statistical distributions based on the data-driven approach.

5.2. General ℓ^p -Norm Uncertainty Set

In this section, we generalize the ellipsoidal uncertainty set to the general ℓ^p -norm uncertainty set while the optimality of linear contracts remains. According to Bertsimas and Brown (2009), an ℓ^p -norm uncertainty set can be a good approximation for the firm when evaluating uncertainty (see Remark 3).

Specifically, we assume that the firm knows that the uncertain parameter vector $\mathbf{a}$ belongs to the following uncertainty set:

$$\mathbb{U}_p = \{(\bar{a}_1, \dots, \bar{a}_n) + (\delta_1 u_1, \dots, \delta_n u_n) \mid \|\mathbf{u}\|_p \leq 1\}, \quad (31)$$

where $p > 1$ and $\|\mathbf{x}\|_p = (\sum_{i=1}^n |x_i|^p)^{\frac{1}{p}}$. Notice that when $p = 2$, $\mathbb{U}_2$ is an ellipsoid.

Then, the firm's optimal robust contract design problem is provided as follows:

$$\begin{aligned}
 \pi(e_h) = &\max_{w_i, i=0, 1, \dots, n} \min_{\mathbf{a} \in \mathbb{U}_p} \sum_{i=1}^n a_i e_h [y_i - y_0 - (w_i - w_0)] + y_0 - w_0, \quad (32) \\
 \text{subject to } &\sum_{i=1}^n a_i (w_i - w_0) \geq \Delta c, \forall \mathbf{a} \in \mathbb{U}_p, \quad (33) \\
 &w_i \geq c(e_h), \forall i = 0, 1, \dots, n. \quad (34)
 \end{aligned}$$

We present the following lemma that is useful to simplify this program.

Lemma 8 (Bertsimas et al. 2004). Consider the following minimization problem:

$$U_p(\mathbf{x}) = \min_{\mathbf{a} \in \mathbb{U}_p} \langle \mathbf{a}, \mathbf{x} \rangle.$$

Then, $U_p(\mathbf{x}) = \sum_{i=1}^n a_i x_i - (\sum_{i=1}^n |\delta_i x_i|^q)^{\frac{1}{q}}$, where $\frac{1}{q} + \frac{1}{p} = 1$. Moreover, $U(\mathbf{x})$ is concave in $\mathbf{x}$.

Similar to Lemma 2, Lemma 8 characterizes the equivalent surrogate utility under the ℓ^p -norm uncertainty set. Following the logic in the basic model, we can show that a linear contract and only a linear contract is optimal under the general ℓ^p -norm ($p > 1$) uncertainty set.

Proposition 6. Under the general ℓ^p -norm uncertainty set, the linear contract is uniquely optimal, that is, $w^*(Y) = \alpha_p^* Y - \alpha_p^* y_0 + c(e_h)$, where the base payment is $c(e_h) - \alpha_p^* y_0$, the

bonus under output Y is $\alpha_p^* Y$ for all $Y = y_i$ for $i = 1, \dots, n$, and the optimal commission rate is

$$\alpha_p^* = \frac{\Delta c}{\sum_{i=1}^n \bar{a}_i (y_i - y_0) - \left(\sum_{i=1}^n |\delta_i (y_i - y_0)|^q \right)^{\frac{1}{q}}}. \quad (35)$$

Remark 10. It is worth noting that assuming $p = 1$, that is, a polyhedral uncertainty set $\mathbb{U}_1$, seems unreasonable in our problem as for any $\mathbf{a} \in \mathbb{U}_1$, every entry a_i must be independent of each other. Recall that $\mathbf{a}$ represents the probability function, meaning the point mass function should be correlated for different outcomes.

Finally, we note that we can also incorporate the ℓ^p -norm uncertainty set into the general output distribution while the optimality of linear contracts remains.

5.3. Multiple Effort Levels

In this section, we show how to generalize the basic model to the case with multiple effort levels. To illustrate the main idea while keeping our presentation simple, we consider the case with three efforts: $e \in \{e_l, e_m, e_h\}$ such that $e_l < e_m < e_h$ with $c(e)$ increasing and convex in e . The case with generic m ($m \geq 3$) possible efforts can be treated analogously.

Define $\Delta c_m = \frac{c(e_m) - c(e_l)}{e_m - e_l}$ and $\Delta c_h = \frac{c(e_h) - c(e_m)}{e_h - e_m}$. Then, $\Delta c_m \leq \Delta c_h$ as $c(e)$ is convex. Suppose that the firm would like to induce the medium effort e_m from the agent, then the firm's optimal robust contract design problem can be formulated as follows:

$$\pi(e_m) = \max_{w_i, i=0,1,\dots,n} \min_{\mathbf{a} \in \mathbb{U}} \sum_{i=1}^n a_i e_m [y_i - y_0 - (w_i - w_0)] + y_0 - w_0, \quad (36)$$

$$\text{subject to } \sum_{i=1}^n a_i (w_i - w_0) \leq \Delta c_h, \forall \mathbf{a} \in \mathbb{U}, \quad (37)$$

$$\sum_{i=1}^n a_i (w_i - w_0) \geq \Delta c_m, \forall \mathbf{a} \in \mathbb{U}, \quad (38)$$

$$w_i \geq c(e_m), \forall i = 0, 1, \dots, n. \quad (39)$$

Based on an argument similar to that in the basic model, we can show that a linear contract and only a linear contract is optimal, which is summarized as follows.

Proposition 7. The linear contract is uniquely optimal to induce the medium effort e_m , that is, $w^*(Y) = \alpha_m^* Y - \alpha_m^* y_0 + c(e_m)$, where the base compensation is $-\alpha_m^* y_0 + c(e_m)$, the bonus under output Y is $\alpha_m^* Y$ for all $Y = y_i$ for $i = 1, \dots, n$, and the optimal commission rate is

$$\alpha_m^* = \frac{\Delta c_m}{\sum_{i=1}^n \bar{a}_i (y_i - y_0) - \sqrt{\sum_{i=1}^n \delta_i^2 (y_i - y_0)^2}}. \quad (40)$$

Substituting Δc_h into α_m^* for Δc_m , we obtain the firm's optimal robust contract to induce the highest effort e_h

from the agent. Comparing two cases yields that to induce a lower effort, for example, e_m compared with e_h , the firm only needs to offer a less-powered contract for the agent, that is, $\alpha_m^* < \alpha_h^*$. This is intuitive in that as exerting effort is costly for the agent, the agent would never exert higher efforts than the targeted one if the firm does not provide enough incentives for those higher efforts.

5.4. Mean-Variance Agent Utility

We show that a piecewise linear contract is uniquely optimal when the agent has a piecewise linear concave utility. In this section, we show that a linear contract is indeed optimal when the risk-averse agent has a mean-variance utility.

To illustrate the main idea, we use the following three-point distribution:

$$Y = \begin{cases} y_2, & \text{w.p. } a_2 e \\ y_1, & \text{w.p. } a_1 e \\ y_0, & \text{w.p. } 1 - a_1 e - a_2 e. \end{cases}$$

where $e \in \{e_h, e_l\}$. We assume the agent has the following mean-variance utility given the compensation plan and his/her own effort:

$$u(w_2, w_1, w_0, e) = w_2 a_2 e + w_1 a_1 e + (1 - (a_1 + a_2)e) w_0 - \gamma e \sqrt{(w_2 - w_0, w_1 - w_0)^T Q (w_2 - w_0, w_1 - w_0)},$$

where $Q = \begin{bmatrix} q_1 & q_3 \\ q_3 & q_2 \end{bmatrix}$ is a positive semidefinite symmetric matrix ($\mathbf{x}^T$ denotes the transpose of a vector $\mathbf{x}$). The matrix Q measures the variability or the risk of a_1, a_2 , whereas γ captures the degree of risk aversion. The term $\gamma e \sqrt{(w_2 - w_0, w_1 - w_0)^T Q (w_2 - w_0, w_1 - w_0)}$ is the risk loss term, and w_0 is the minimum guaranteed payment when there is no risk.

Remark 11. The mean-variance utility is not an expected utility and is extensively used in the finance literature. Moreover, Markowitz (2014) argues that the mean-variance utility approximately maximizes expected utility for a variety of concave (risk-averse) utility functions.

In contrast to the basic model, we assume that (a_2, a_1) is in the following ellipsoid:

$$\mathbb{U}_{MV} = \left\{ (a_2, a_1) \mid ((\bar{a}_2, \bar{a}_1) - (a_2, a_1))^T Q^{-1} ((\bar{a}_2, \bar{a}_1) - (a_2, a_1)) \leq \delta^2 \right\}. \quad (41)$$

This type of uncertainty set is adopted in Garlappi et al. (2007) and Zhu et al. (2009) for robust portfolio selection. We impose the following assumption.

Assumption 3. We have (i) $\bar{a}_i > \delta_i$ for $i = 1, 2$ and $1 - (\bar{a}_1 + \bar{a}_2)e_h > e_h \sqrt{\delta_1^2 + \delta_2^2}$; and (ii) $\bar{a}_2 - (\gamma + \delta)\sqrt{q_1} > 0$ and $\bar{a}_1 - (\gamma + \delta)\sqrt{q_2} > 0$.

Part (i) in Assumption 3 ensures the probability for each output realization is positive, whereas part (ii) guarantees the agent still prefers higher payment even under the worst-case criterion. We can formulate the firm's robust contract design problem to induce high effort e_h as follows:

$$\max_{w_2, w_1, w_0} \min_{(a_2, a_1) \in \mathbb{U}_{MV}} [(y_2 - w_2)a_2e_h + (y_1 - w_1)a_1e_h + (y_0 - w_0)(1 - (a_1 + a_2)e_h)], \quad (42)$$

$$\text{subject to } u(w_2, w_1, w_0, e_h) - c(e_h) \geq u(w_2, w_1, w_0, e_l) - c(e_l), \forall (a_2, a_1) \in \mathbb{U}_{MV}, \quad (43)$$

$$w_2, w_1, w_0 \geq c(e_h). \quad (44)$$

Notice that when $\gamma = 0$ and $Q = \frac{1}{\delta^2} \begin{bmatrix} \delta_2^2 & 0 \\ 0 & \delta_1^2 \end{bmatrix}$, then $\mathbb{U}_{MV}$ is the same as that in the basic model when $n = 2$, that is, the basic model is a special case of the robust contract design problem with mean-variance agent utility.

Proposition 8. Under Assumption 3, the only optimal robust contract is a linear contract: $w^*(Y) = \alpha_{MV}^* Y - \alpha_{MV}^* L + c(e_h)$, where the base compensation $c(e_h) - \alpha_{MV}^* L$, the bonus under output Y is $\alpha_{MV}^* Y$ for $Y = y_0, y_1, y_2$, and the optimal commission rate is

$$\alpha_{MV}^* = \frac{\Delta c}{\bar{a}_2(y_2 - y_0) + \bar{a}_1(y_1 - y_0) - [\gamma + \delta]} \times \sqrt{(y_2 - y_0, y_1 - y_0)^T Q (y_2 - y_0, y_1 - y_0)}. \quad (45)$$

In particular, the optimal commission rate α_{MV}^* is increasing in δ, γ and decreasing in y_2, y_1 .

Proposition 8 shows that the optimality of linear contracts can go beyond the agent's risk-neutral utility. Recall that with a piecewise linear risk-averse utility, the optimal robust contract is a piecewise linear contract. In sharp contrast, with a mean-variance risk-averse agent utility, a linear contract turns to be optimal. This is due to the intrinsic connection between the ellipsoidal uncertainty set and the mean-variance utility (see Remark 3). To see this, notice that the agent's surrogate utility induced by the ellipsoidal uncertainty set has the same functional form as the mean-variance utility. As a result, under the worst-case criterion, both the firm and the agent still have homogenous uncertainty-adjusted terms in their surrogate utilities, thus the sharing scheme in the basic model, that is, a linear contract, still works in this scenario. This logic applies to cases where the firm has a mean-variance risk-averse utility and both parties have risk-averse utilities as $U_p(x)$ characterized in Lemma 8. In these scenarios, with some modification to part (i) of

Assumption 3, the optimality of the linear contract remains. Moreover, it is intuitive that if the agent is more risk averse, the firm has to compensate more to motivate the agent, that is, α_{MV}^* is increasing in γ .

6. Concluding Remarks

In this paper, we introduced model uncertainty in contract designs under moral hazard. Specifically, we assumed that the parameters for the underlying effort-contingent distribution of the random output are in some uncertainty set, for example, an ellipsoid. To tackle the parameter uncertainty issue, we proposed the notion of robust contract design: given a contract, the firm (the principal) evaluates its worst-case performance; a contract is incentive compatible for any possible parameters in the uncertainty set (robust incentive compatibility); and the firm would like to identify the contract that maximizes its worst-case performance subject to the robust incentive compatibility. The firm is financially risk neutral and the agent has limited liability. The robust contract design seems challenging. However, we can transform the robust contract design problem into a simple optimization problem based on robust optimization. We found that for a financially risk-neutral agent, a linear contract is optimal: The firm pays the agent a fixed base payment and a fixed share or commission rate of the realized output. In contrast to the existing literature, we showed that the linear contract is the only type of contract that is robust to the parameter uncertainty. Other contract forms, including nonlinear contracts, are neither optimal nor robust to the parameter uncertainty.

We also found that when the agent has a piecewise linear concave utility, the only optimal robust contract is a piecewise linear contract that consists of progressive fixed payments and linear rewards with progressive commission rates. Thus, from the perspective of robustness, we provide a new explanation for why linear contracts as well as piecewise linear contracts are so popular in practice. Moreover, we analyzed the trade-off between robustness and the worst-case performance and provided guidelines on how to choose the uncertainty set.

We further showed that our results are robust to a variety of settings. We first generalized our results to the case where the firm has ambiguity over a general discrete effort-contingent distribution of the random output. We showed that under the general output distribution with MLRP, a generalized linear contract is uniquely optimal. We also showed that our results are robust to the cases with general ℓ^p -norm uncertainty sets, multiple effort levels, and the risk-averse agent with a mean-variance utility. Finally, we discuss how the participation constraint and inconsistent

information sets affect the optimal robust contract in Online Appendix C.

Our paper presents a first cut for robust contract designs under moral hazard. Our modeling framework is flexible and we provided tractable tools that yield simple solutions. Hence, our modeling approach may allow us to analyze robust contract designs in other agency settings.

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Endnote

¹ We thank the associate editor for this example.

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